LLM ENGINEERING

Vector Databases Visual Guide

Search by meaning at scale:
ANN, HNSW, IVF, and vector DB architecture.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The ANN Problem

Exact search doesn't scale; approximate does.

Brute force

compare query to ALL 100M vectors.

Accurate, impossibly slow.

ANN

99% of accuracy at 1% of the cost.

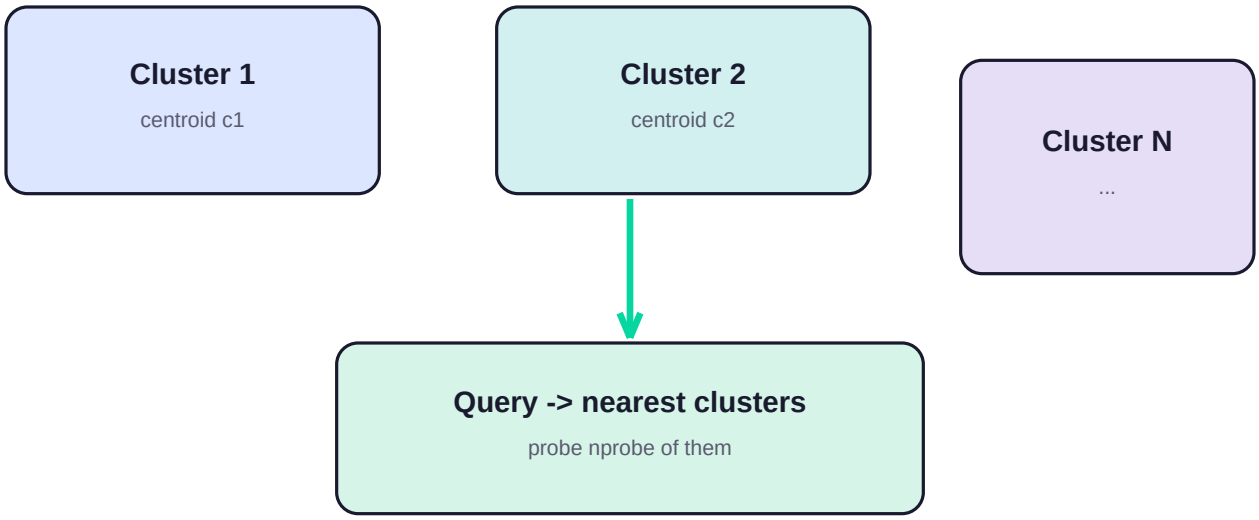
The recall/latency tradeoff.

ANN = Approximate Nearest Neighbor. Nobody needs the mathematically closest vector — the 10th-closest is just as useful.

STEP 2/6

IVF: Clustering the Space

Search the right neighborhood, not the whole city.



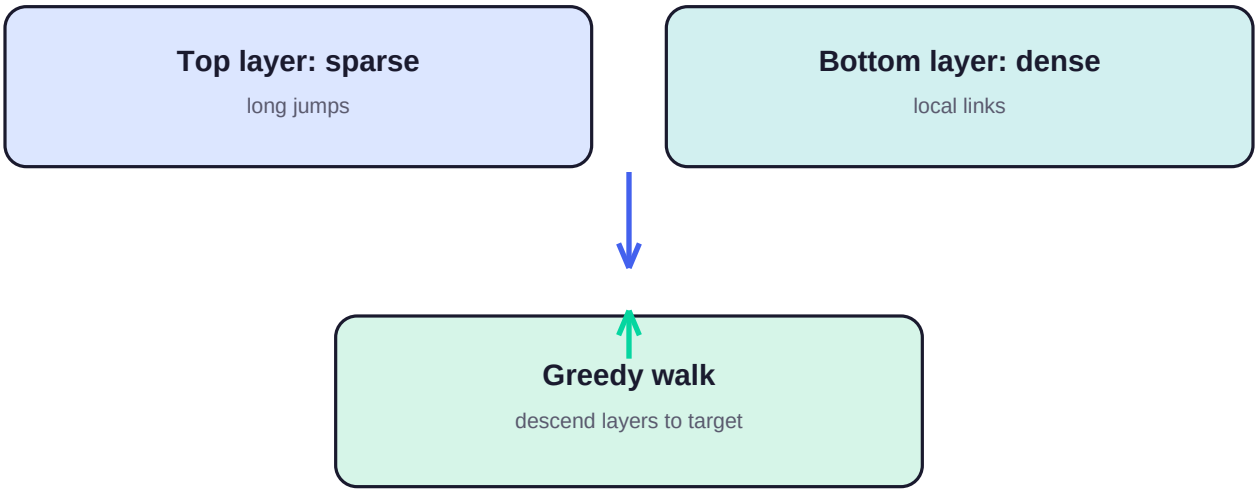
Train k-means centroids; assign each vector to a cluster. Query only the closest nprobe clusters.

Tuning: more clusters = finer partitions; higher nprobe = better recall, slower search.

STEP 3/6

HNSW: The Graph Index

Navigate a layered highway of vectors.



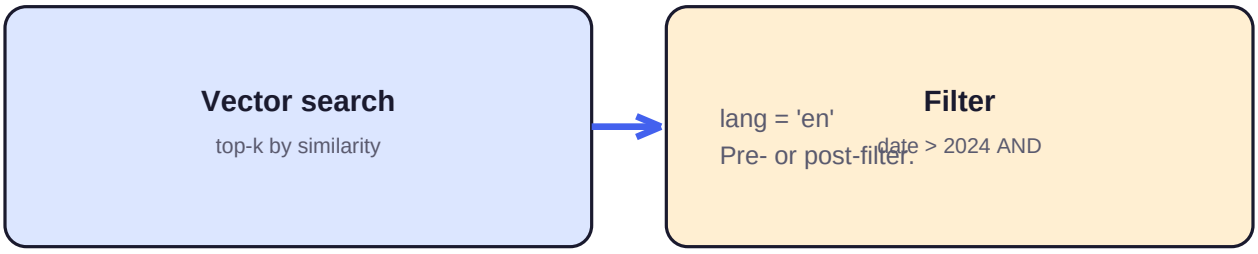
Hierarchical Navigable Small World: enter at the top, greedily walk toward the query, refine downward.

HNSW is the default high-recall index. Cost: significant RAM — the graph lives in memory.

STEP 4/6

Metadata Filtering

Vectors plus structured filters.

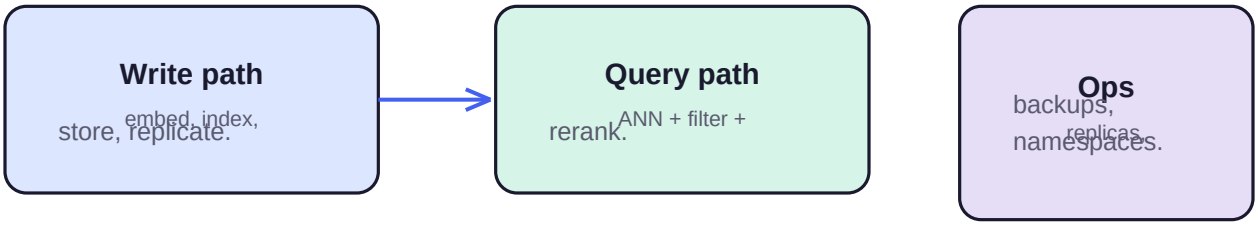


Pre-filtering is exact but can starve ANN; post-filtering is fast but may return too few. Hybrid indexes balance both.

STEP 5/6

Vector DB Architecture

What Pinecone / Weaviate / Qdrant provide.



Managed (Pinecone) vs self-hosted (Weaviate, Qdrant, Milvus): tradeoff is operational burden against control and cost.

STEP 6/6

Tuning Recall vs Latency

The knobs that matter.

- HNSW: ef_search higher = better recall, slower queries. Build once with high ef_construction.
- IVF: nprobe higher = better recall, more clusters scanned.
- Quantize vectors (PQ/SQ) to fit bigger indexes in RAM — small recall cost.
- Measure recall@k against brute-force ground truth on YOUR data, not vendor benchmarks.
- Set an SLO (p99 < 50ms at recall 0.95?) then tune the index to meet it.

STEP ✓I✓

Key Takeaways

Vector Databases — Remember

- ANN trades a little accuracy for massive speed — the 10th-closest vector is fine.
- IVF clusters the space and probes nearby clusters; HNSW walks a layered graph.
- Combine vector search with metadata filters for real applications.
- Managed vs self-hosted is an ops-burden vs control tradeoff.
- Tune ef_search/nprobe against recall@k measured on your own data.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev